

A New Optical Fiber-Based Sensor for Concentrated Solar Flux Monitoring in Concentrated Solar Power Plants

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Abstract

This paper presents the proof of concept of a novel optical fiber-based sensor to measure concentrated solar flux up to 1 MW/m² in Concentrated Solar Power (CSP) applications. The proposal constitutes an efficient, passive, robust and low-maintenance alternative to traditional radiometers. ©2026 The Author(s)

Introduction

The profitability and efficiency of Concentrated Solar Thermal (CST) and Concentrated Solar Power (CSP) systems rely heavily on the accurate and continuous monitoring of solar flux distribution. Currently, state-of-the-art monitoring relies on radiometers and camera-based techniques. While functional, these systems operate under strict requirements and demand recurrent maintenance, creating an undesirable dependence in harsh operational environments. Moreover, their inclusion in the reception area can occasionally be limited due to their size and dimensions. This paper presents a novel, passive, and minimum-invasive optical fiber-based approach. By utilizing *ad-hoc* optical fibers acting as direct radiation collectors and coupling them with thermal photodetectors with their respective optical power meters, a flexible alternative to measure concentrated solar flux is proposed. This work focuses on the optical design of the solution, the material constraints of the fibers, and the mathematical data treatment required for the solar flux estimation.

Furthermore, this alternative holds special practical interest within the framework of the European project ASTERix-CAESar. In this context, developing a minimally invasive and highly robust sensor is crucial to successfully monitor receivers under extreme conditions, with the goal of reaching and validating concentrated solar flux measurements of up to 1 MW/m².

System Description

In CSP applications, optical components must withstand extreme thermal conditions. To achieve this, special optical fibers were designed and manufactured by engionic Femto Gratings GmbH (Goslar, Germany), featuring temperature-resistant gold coatings housed inside a

stainless-steel capillary. These optical fibers are responsible for receiving and coupling the incident light coming from CSP fields. To characterize the optical signal, a Thorlabs S401C thermal power sensor was utilized as a photodetector. This device converts coupled light into either current or optical power, depending on the configuration. The most restrictive quality of these sensors tends to be their low sensitivity. This particular model has a resolution of 1 μW, meaning that applications managing power values that do not surpass this magnitude order might be inaccurate and present undesirable steps. The values given by the thermal sensor are extracted via a power meter and subsequently processed within the Data Acquisition (DAQ) system. This setup enables the measurement of the concentrated flux generated in the solar field. Nevertheless, the following preliminary calculation must be considered in order to obtain the estimated raw solar flux.

$$\phi_{raw} = \frac{P_{opt}}{A_{eff}} \quad (1)$$

$\phi_{raw}(\lambda_i)$ is derived from the measured optical power, $P(\lambda_i)$, divided by the effective capturing area of the optical fiber, A_{eff} . For the multimode optical fibers utilized in this high-irradiance architecture, this effective collection area corresponds directly to the core area, calculated as $A_{eff} = \pi \times a^2$, where a corresponds to the radius of the fiber core. By applying this equation, the reduced optical power that is coupled into the fiber can conserve its nature and represent the real incident flux.

Proof of Concept

An experimental campaign was carried out at the High Flux Solar Simulator (HFSS) KIRAN42 (IMDEA Energy, Móstoles, Spain) to test the functionality of the complete system. This facility, based on Xenon arc lamps, is able to achieve a maximum peak flux around 3.5 MW/m^2 , making it a suitable environment for emulating real working conditions commonly met in solar receivers. The positioning of the optical sensing heads was structurally supported by an Inconel Alloy 625 flat plate coated with Pyromark 2500, which also served as a coordinates mapping reference. As observed in Fig. 1, the fiber tips were exposed directly to the high irradiance environment. After collecting the incident light, the optical fibers passed through the internal cavity of the mounting plate via 4-mm apertures. As shown in Fig. 2, they exited from the rear to route the captured optical power directly to the measurement instruments.

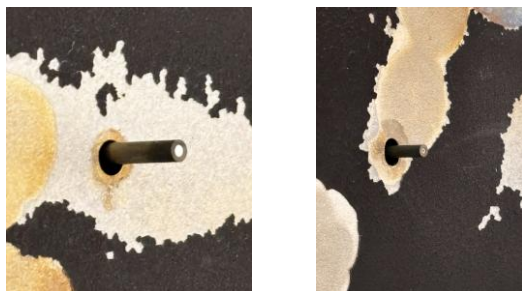


Fig. 1: Close-up views of the specialized gold-coated optical fiber tip protruding from the protective plate, acting as the primary radiation collector.



Fig. 2: Backside of the mounting plate showing the routing of the embedded optical fibers.

This receptor prototype was positioned in a motorized platform that allowed the fibers to change their location to the centre of the focus point of the Xenon lamps. After the instrumentation, different experiments were prepared to assess the resilience of the fibers to the variation of the incident concentrated flux. Principally, the main interest relied on the angle of incidence of

the light beam. Accordingly, the central and lateral lamps were used to simulate direct and oblique irradiance.

For the initial case of normal irradiance (central lamp on), the system showed excellent long-term stability, reduced rippling and high accuracy for a reference of 312 kW/m^2 (see Fig. 4a), although its time response was slower than the traditional radiometers. Moreover, the flux magnitude demonstrated no degradation along the experiment, without signal drop-offs nor decreasing inertia.



Fig. 3: High Flux Solar Simulator KIRAN-42 (IMDEA Energy, Mostoles, Spain) with the lateral lamp on.

Secondly, a lateral lamp was powered on instead of the central lamp (see Fig. 3). In this scenario, the results were drastically different. As shown in Fig. 4b, the concentrated flux measurement suffers from various phenomena that alter its behaviour and results. The maximum registered concentrated flux is almost thirty times less than the direct exposure case. Additionally, the stability and reduced rippling of the first case have been replaced by a decreasing tendency and the presence of pronounced peaks and valleys (high rippling).

Concerning this issue, a critical factor is currently under investigation, which exerts a dominant influence on the measurement of concentrated solar flux. This is the angular acceptance of the optical fiber. Optical fibers have an intrinsic limitation in regard to their field of view (FOV). Unlike the measurement of Direct Normal Irradiance (DNI), where the incident radiation is constrained to a narrow solid angle ($\sim 0.5^\circ$) centred on the solar disk, concentrated solar flux involves radiation incident from a significantly wider angular range, determined by the specific optical geometry of the heliostat field or the parabolic

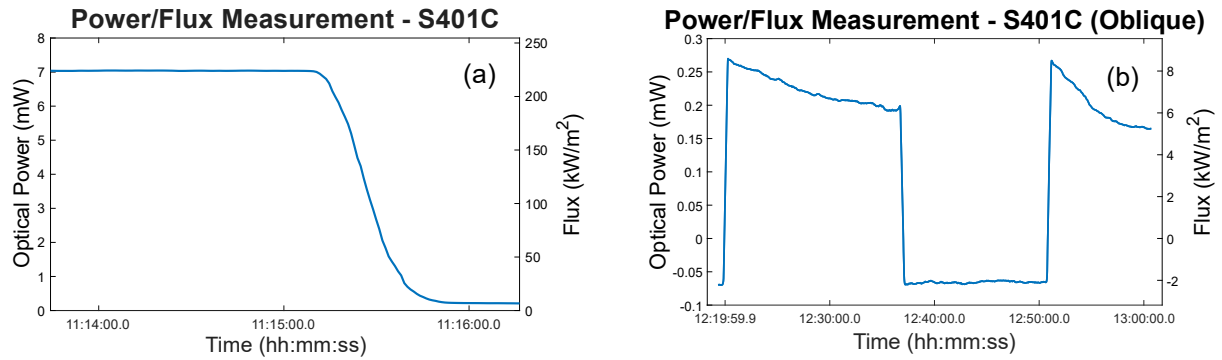


Fig. 4: (a) Optical Power Measurement for a 312 kW/m² flux reference, using the S401C photodetector – Direct Exposure; (b) Optical Power Measurement for a 357 kW/m² flux reference, using the S401C photodetector – Oblique Exposure.

concentrator. The fiber Numerical Aperture (NA) defines a specific half-acceptance angle, $\theta_a = \arcsin(NA)$, within which light can be effectively coupled into the core. For the standard multi-mode fibers used in this study in a CSP receiver environment, rays incident outside its critical cone are not coupled, leading to a notable and systematic reduction in the measured flux values—an effect that is considerably more pronounced than in standard radiometric applications.

Furthermore, measurements appear to have a strong dependence on the temperature of the fiber tip. When directly exposed, the fiber does not suffer from radiation as almost the totality of the light is coupled into the fiber. However, when the incidence is oblique, a noticeable percentage of the incoming light impacts in the external part of the fiber. Although no damage is visible, the results suggest that the fiber suffers degradation and loses coupling capacity. When the lamp is powered off, the fiber temperature decreases progressively, until it returns to its original state. Once the incoming flux reaches the fiber and maintains, the process repeats.

Conclusions

This study demonstrates the feasibility of using optical fibers with thermal power sensors for direct solar flux measurements in CSP environments. While the low sensitivity that characterizes these photodetectors creates a considerable limitation in terms of minimum power required, CSP high concentrated flux eliminates its importance by providing enough margin to improve the accuracy and minimize the resolution impact. Future optical designs will focus on widening the system's field of view and mitigating the temperature dependence of the measurements.

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